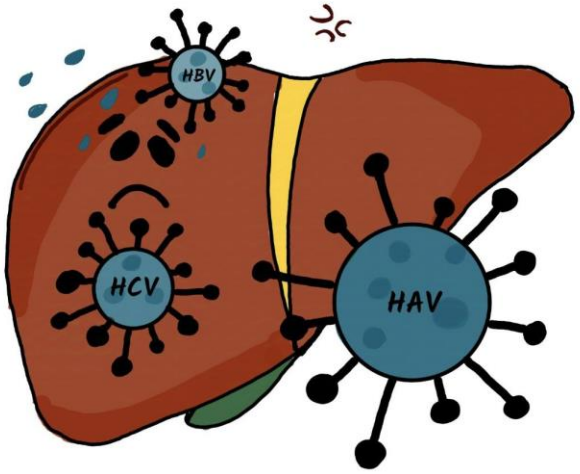
Pediatrics

- Dermatology
- Allergy
- Hepatology
- Liver tumor & Other important pediatric tumors
- Child Abuse



ملاحظة مهمة شابتر oncology مشروح كالآتي: -

- Leukemia and lymphoma (blood chapter)
- Brain tumors (CNS chapter)
- Wilms tumor and neuroblastoma (Renal chapter)

د. عبدالرحمن الجعكي

2025_2024

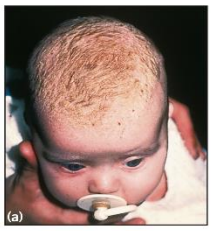
Pediatric Dermatology

Infantile Seborrheic dermatitis

- This eruption of unknown cause presents in the first 3 months of life .
- It starts on the scalp as an erythematous scaly eruption. The scales form a thick yellow adherent layer, commonly called cradle cap
- The scaly rash may spread to the face, behind the ears, and then extend to the flexures and napkin area
- In contrast to atopic eczema, it is not itchy, and the child is unperturbed by it.
- It is associated with an increased risk of subsequently developing atopic eczema.

Rx

- Reassures parents it usually resolved spontaneously.
- Mild case emollients (Emollient with low sulfur & Sali cyclic acid daily they washed off)
- Mild topical steroids & Antibacterial or Antifungal in severe cases.



Atopic eczema (Atopic dermatitis)

- Prevalence ≃20% in UK
- Age in 1st year of life & disease of childhood (uncommon in the first 2 months)
- Associated with other atopic diseases (+ F/H of atopy) Asthma – heavy fever and allergic rhinitis
- ≃ 1/3 of child with eczema develops br asthma
- Breast feeding: Delay the onset of eczema



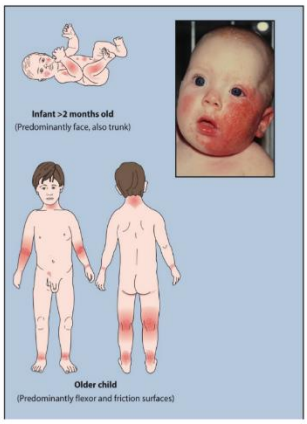
Prognosis:

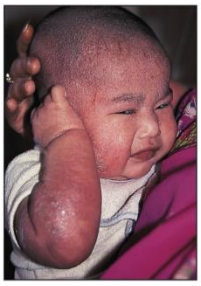
Atopic eczema is mainly a disease of childhood, being most severe and troublesome in the first year of life.

- 50% resolved by 12 years of age
- 75% resolved by 16 years of age

Clinical picture: Clinically, rash associated with itching (pruritus) in all age , scratching – excoriation , lichenification

- **Site:** Infant > 2m (face also trunk)
- Older child (Flexor & friction surfaces)





Atopic dermatitis. Inflamed skin worsened by rubbing/scratching. Itch is the key clinical feature in eczema at all ages, leading to an 'itch–scratch–itch' cycle.

Complication:

1. Flare up – causes of exacerbation

   - Bacterial infection eg staph – strep
   - Viral infection eg HSV
   - Ingestion of an allergen
   - Contact to irritant or allergen
   - Environment heat humidity
   - Change or decrease in medication
   - Psychological stress
   - Unexplained

2. Skin infection (staph) eczema or (HSV) : rare and cause eczema herpeticum.

3. Regional LN

4. Psychological impact.

Treatment:

1. Ovoid irritant & tigger factors :avoid soap and biological detergents. Clothing next to the skin should be of pure cotton.

2. Emollients (mainstay of Rx), moisturizing and softening of skin and ointment better than cream in dry skin

3. Topical CS – mild potent steroid (H.C 1%) and moderate potent steroid for acute exacerbation

4. Immunomodulators (>2y) not controlled by steroid: as tacrolimus ointment

5. Antibiotic – Antiviral – Antihistamine

   - If severe back infection – systemic antibiotic
   - If Eczema herpeticum – acyclovir
   - If itching – 2nd generation antihistamine

6. Dietary elimination : extensive HA formula .

7. Psychological support (both child and family)

Differential diagnosis of itchy skin rash

MCQ

1. Atopic eczema
2. Chickenpox
3. Urticaria/allergic reactions
4. Contact dermatitis
5. Insect bites
6. Scabies
7. Fungal infections
8. Pityriasis rosea

Nappy Rash

مهم جدا في البيبر و الكلينك

Nappy rash is common in infants.

Differential diagnosis

1. irritant dermatitis (most common)

- Infrequent change of napkin or if the infant had diarrhea
- The rash is due to the irritant effect of urine on the skin of susceptible infants.
- Urea–splitting organisms in faeces increase the alkalinity and likelihood of a rash.
- Rash erythematous – scaled – ulcer formation,,
- Flexures (skin fold) are spared.
- Rx: Zinc oxide cream & emollient , mild topical steroid in severe cases



Figure 25.6 Napkin rash due to *Candida* infection. The skin flexures are involved and there are satellite pustules visible.

2. Candidal (monilial) dermatitis

- Caused by candida albicans
- Skin folds involved + satellite lesions
- ± Oral thrush
- Rx: topical antifungal

3. Seborrheic dermatitis

4. Atopic eczema

5. Acrodermatitis enteropathica (zinc deficiency)

6. Langerhans cell histiocytosis

7. Wiskott – Aldrich syndrome

Viral warts:	Molluscum contagiosum:
• Human papillomavirus • Common in children (Fingers and toes) • Resolve spontaneously • Rx: indicated if painful or cosmetic problem with Cryotherapy	• Pox virus • Papule with central umbilication • Single or multiple • Highly Contagious • Resolve spontaneously (yr) • Rx: cryotherapy old child (not face)
	 

Scabies: Mcq

- Caused by parasite (mite) sarcoptes scabiei (IP : 2–6 week)
- Pathognomic feature (burrow)

Clinical picture

- Severe itching– more at night and increase warm condition
- Papules and vesicles involve the skin between the fingers and toes, axillae, flexor aspects of the wrists, belt line, and around the nipples, penis and buttocks.
- In infants and young children, the distribution often includes the palms soles, and trunk
- All family members may have itching.

Diagnosis:

Clinically & Skin scraping for microscopic examination (to identify mite, eggs)

Treatment

- All family members should be Rx.
- Permethrin cream (5%)
- Benzyl benzoate (25%)
- Malathion lotion (0.5%)

Pediculosis:

- Pediculosis capitis (head lice) is the most common type .
- Common problem among primary school children

C/P

- Itching of scalp and nape
- 2 year bacterial infection (misdiagnosed as impetigo)
- Lymphadenopathy (Occipital)

Rx

- Malathion 0.5% left overnight
- Dimeticone 4% lotion
- Wet combing with a fine– tooth comb to remove live lice (bug–busting) every 3–4 days for at least 2 weeks is a useful and safe physical treatment.k
- Rx repeated > 1 week.

Pityriasis rosea

- This acute, benign self–limiting condition is thought to be of viral origin.
- It usually begins with a single round or oval scaly macule, the herald patch, 2–5cm in diameter, on the trunk, upper arm, neck, or thigh.
- After a few days, numerous smaller dull pink macules develop on the trunk, upper arms, and thighs.
- The rash tends to follow the line of the ribs posteriorly, described as the 'fir tree pattern'.
- Sometimes the lesions are itchy. No treatment is required and the rash resolves within 4–6 weeks.

D/D of Skin rash and fever

1–Maculopapular rash	2–Vesicular, bullous, pustular rash	3–Petechial, purpuric rash.
Viral • Human herpesvirus–6 (HHV–6) or HHV–7 (roseola infantum) • Enterovirus rash • Parvovirus (slapped cheek) – usually school age • Measles – uncommon if immunized • Rubella – uncommon if immunized **Bacterial** • Scarlet fever (group A streptococcus) ,Toxic shock syndrome • Rheumatic fever • Typhoid fever) :rose spots • Lyme disease: erythema migrans **Other** • Kawasaki disease • Systemic onset juvenile idiopathic arthritis	**Viral** • Chickenpox • Herpes zoster (shingles) • Herpes simplex virus • Coxsackie virus – hand, foot and mouth **Bacterial** • Impetigo • Boils • Staphylococcal bullous impetigo • Staphylococcal scalded skin **Other** • Erythema multiforme/ Stevens– Johnson syndrome • toxic epidermal necrolysis	• Meningococcal, other bacterial sepsis • Henoch–Schönlein purpura • Thrombocytopenia as ITP. • Infective endocarditis • Enteroviruses, adenoviruses. • Vasculitis • Malaria • DIC.

Pediatric Allergy

Allergens:

- Antigens are able to induce IgE production which must lead to type I hypersensitivity response upon subsequent exposure to the same protein.
- They includes food allergy – Eczema – Allergic – Rhinitis – Asthma
- In UK 40% of children have allergic rhinitis – asthma and eczema and 6% develop food allergy
- They are the most common chronic disease and school absence

Anaphylaxis:

Serious allergic reaction that is rapid onset and may cause death

Allergy:

Hypersensitivity reaction initiated by specific immunological mechanism, it can be IgE mediated (peanut allergy) and non IgE mediated (Celiac disease)

Atopy:

Personal and/or familial tendency to produce IgE Ab in response to ordinary exposure to potential allergen strongly associated with Asthma – Allergic Rhinitis – Conj – Eczema and food allergy

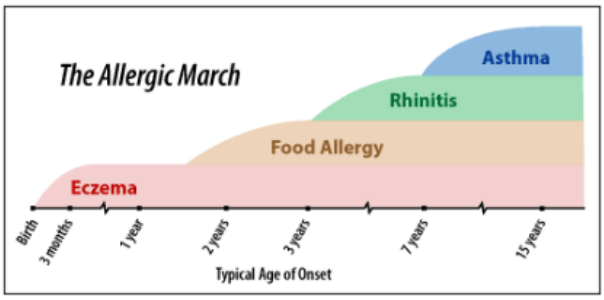
Allergic March:

- Allergic children develop other allergic diseases at different ages
    - Eczema – food allergy – infancy
    - Allergic Rhinitis and conj – preschool
    - Then asthma – preschool and 1yr school

Up to 80% of children with asthma have allergic rhinitis

Prevention of Allergic disease:

- Exclusive breast feeding
- Use of HA formula
- Use of probiotic for eczema
- Medication (immunotherapy)
- Nutritional supplement (Omega 3, Vit D and Antioxidant)



Allergic Rhinitis

Risk factors

- F/H of atopy
- Early introduction of food or formula
- Maternal heavy smoking

Clinical picture

- Sneezing – nasal pruritus – nasal congestion – rhinorrhea – mouth breathing – sleep disturbances
- Dennie – morgan lines (under the lower eyelids)
- Allergic shiners (Bluish discoloration infra orbital)
- Allergic salute : nasal transverse crease due to rubbing
- Eye redness – puffiness – itching
- Swelling of the nasal mucosa and turbinates and nasal polyps with mucosal pallor

Treatment



- Allergen avoidance
- Intranasal antihistamine (>5y)
- Intranasal steroid** (budesonide)
- Oral antihistamine (non sedating).
- Nasal decongestant (not more 7–10 days) : rhinitis medicamentosa !!
- Allergen immunotherapy (S.C) and sublingual (used also in asthma eczema – SC, sublingual for 3–5 years)
- Systemic steroid should be avoided
- Leukotriene antagonist (montelukast)

Urticaria

- ✓ Urticaria (hives): is characterized by wheal.
- ✓ Papular urticaria:
  - delayed hypersensitivity reaction following bite from flea – animal – mite.
  - site legs most common
  - c/p vesicle papules last week or months may recurrent.





Anaphylaxis in children/adolescents ↓

مهمة جدا اميرجنسي.

- Anaphylaxis is a severe, life–threatening, sudden–onset systemic allergic reaction.
- In children,about two thirds of anaphylaxis is caused by food allergy. Other triggers such as insect stings or medication are much less common than in adults.
- Most paediatric anaphylaxis occurs in children under 5 years of age, but the majority of fatalities occur in adolescents with allergy to milk followed by peanut and tree nuts. Asthma is an additional risk factor. No trigger is iden– tified in about 20%.
- Reaction is mainly to foods – about 1 in 1000 episodes is fatal.
- Acute management is ABCDE and early administration of intramuscular adrenaline.

"Emergency"

Anaphylaxis

Diagnosis of anaphylaxis – check for:

- Acute onset of illness
- Life-threatening problems:
  - Airway: swelling, hoarseness, stridor
  - Breathing: tachypnoea, wheeze, fatigue, cyanosis, SpO_2 <92%
  - Circulation: pale, cold extremities, weak pulses, low blood pressure
  - Disability: feeling faint, drowsy, coma
- Skin changes – urticaria (usually)

Call for help

- Lie patient flat
- Raise patient's legs

Adrenaline (intramucular)

Every 5 mins until clinical improvement

When specialist equipment available

- Establish airway
- High flow oxygen
- IV fluid challenge
- Chlorphenamine
- Hydrocortisone

Monitor:

- Pulse oximetry
- ECG
- Blood pressure

Intravenous adrenaline infusion if 3 cycles of intramuscular adrenaline fails to stabilize the patient, given under anaesthetic supervision



Viral Hepatitis

- HAV – HBV – HCV – HDV – HEV and HGV
- All are RNA viruses except HBV (DNA)

Mode of transmission

- HAV & HEV :Feco oral
- HBV – HDV & HCV :Parentral and sexual
- HBV :Transplacental – dialysis

- Chronic carrier state :only HBV – HCV & HDV
- Vaccine present :HAV – HBV & HDV
- Treatment available :
  - Interferon for HBV – HCV – HDV
  - Lamivudine in HBV
  - Ribavirin oral in HCV

Clinical differentiation

HAV:

Ip (30D), most common, school age, acute and short course – No carrier state and fulminant hepatitis occurs but rare

HBV:

IP (2–4m), insidious onset and prolonged course, chronic carrier state occurs and fulminant hepatitis may occur.

HCV:

Ip (2m), as B but more prolonged, fulminant hepatitis is rare

HDV:

Ip (2–4m), produces infection only with HB infection more severe but less chronic and more 50–70%.

Acute viral hepatitis

Clinical picture

- Vomiting – nausea – abdominal pain – lethargy – jaundice
- (30–50 of children do not develop)
- Tender hepatomegaly is common
- Splenomegaly ≃ 30%
- Liver transaminases marked by high
- Coagulation is usually normal

Acute liver failure (Fulminant hepatitis)

Is massive hepatic necrosis e⁻ loss of liver function with or wothout hepatic encephalopathy (High MR) 70%.

Caused infection on metabolic disease

C/P: Jaundice – encephalopathy – coagulopathy – Hypoglycemia – Electrolyte disturbances

Complication: Cerebral edema – He – Gastritis –sepsis – Pancreatitis

Diagnosis

- LFT ALT – ASH High (10* – 100*)
- Bil high (early normal)
- PT – PTT prolonged
- Plasma ammonia increases
- RBS – ABG – CT scan

Treatment

- Stabilization 1st ABC
- Correct hypoglycemia
- Prevent sepsis : antibiotic IV
- Prevent Hge : PPI – Vit K
- Prevent central edema : fluid restriction
- Liver transplantation.

Poor prognostic features

- Worsing encephalopathy
- Small liver "Shrinking"
- Progression to coma
- Rising bilirubin with falling transaminase.

Hepatitis A:

- Asymptomatic
- Majority of children have mild illness recover within 2 weeks
- Diagnosis serology (Igm) High
- No Rx – no bed rest – Change of diet is ineffective
- HAV – Vaccin close contact within 2 weeks and traveler to endemic areas

Hepatitis B:

- Infants e^- perinatal HBV are asymptomatic but ≃ 90% become chronic carrier
- Older children may be asymptomatic or classical acute hepatitis most of cases

Resolve spontaneously

1–2% (Fulminant hepatitis)

5–10% (Chronic carrier)

30–50% chronic HBV liver disease

10% may liver cirrhosis – carcinoma

Prevention:

Babies of all HBsAg – positive mothers should receive hepatitis B – immunoglobulin and hepatitis B vaccine

(HB serology inter pretation)

HBs Ag	Anti HBc	HBs Ab	
+	IgM +	–	Acute infection
+	IgG +	–	Chronic infection
–	IgG +	+	Resolved infection
–	–	+	Vaccinated

HBc Ag : viral replication (increase infectivity)

Hepatitis C

- Vertical transmission is the most common cause of HCV transmission in children (increase if coin infection e HIV)
- Acute hepatitis is more but majority become chronic carrier 20–25% progress to cirrhosis & carcinoma
- Rx combination of interferon + ribavirin is successful and sofosbuvir (new) 100% curative.

Complication of chronic liver disease

1–Nutrition:

- Fat malabsorption (R_x by medium chain TG and fat soluble vitamin)
- Protein malabsorption
- Anorexia : R_x by NGT – TPN

2–Encephalopathy:

- In end stage liver disease
- C/P: Irritability – sleepiness – behavior changes – LOC
- Oral lactulose and oral antibiotic (decrease ammonia)

3–Cirrhosis and portal HTN:

- Extensive fibrosis with regenerative modules
- Portal HTN – ascites– splenomegaly – hepato cellular carcinoma may develop.
- Physical sign: Jaundice – Palmar erythema–spider Naevi – malnutrition – Hypotonia – Caput medulla and splenomegaly

4–Esophageal Varices:

- Diagnosed by Upper GI endoscopy
- C/P: upper GI bleeding
- Rx – PRBCs transfusion – H2 blocker – octreotide IV – vasopressin – endoscopic ligation or sclerotherapy – portacaval shunt– liver transplantation

5– Ascites:

Why??

Liver Transplantation

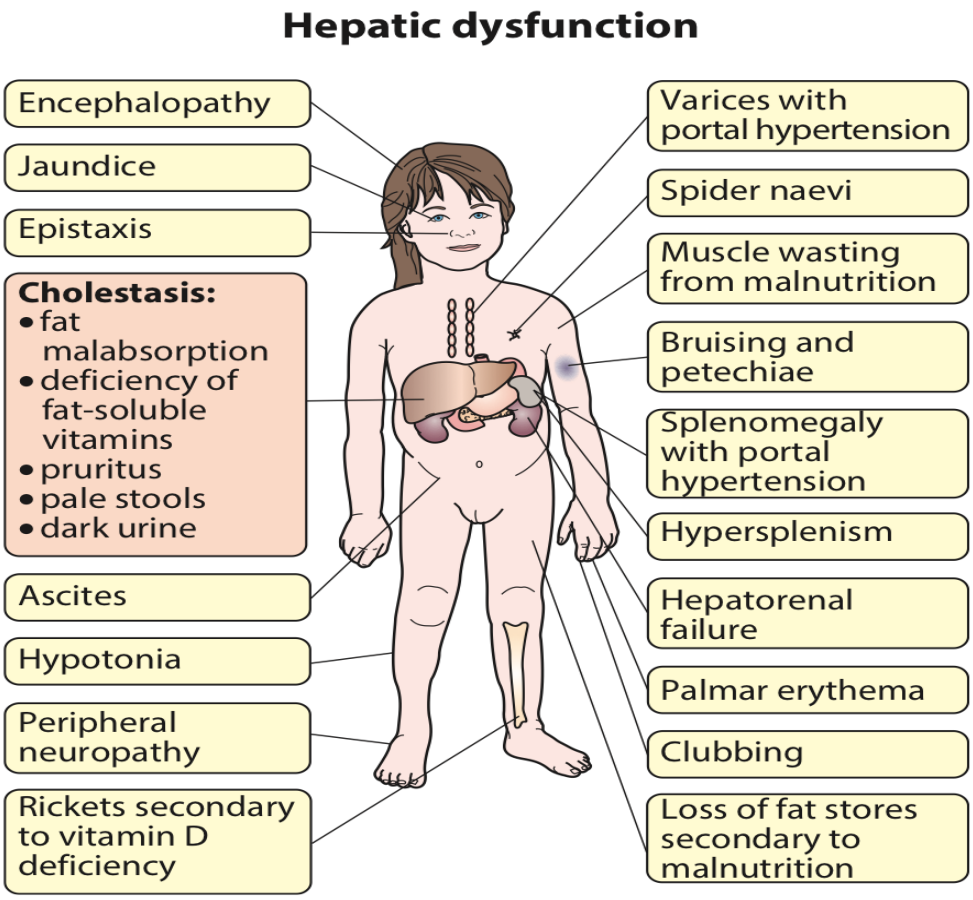
For acute and chronic end stage liver failure

- Low albumin.
- Na retention.
- Renal impairment 2 yr to ATN.

6–Spontaneous bacterial peritonitis:

- C/P : fever –abdominal pain – abdominal tenderness
- Diagnosed by paracentesis
- Rx : Broad spectrum antibiotics

Sign and symptoms of chronic Liver disease (MCQ)



✓ Note.(موضوعbiliary atresia شرح بالتفصيل في شابتر نيونيت.)

- **Cause of acute liver failure:**

Children < 2 years old	Children > 2 years
Infection (most common is herpes simplex)	Seronegative hepatitis
Metabolic disease	Paracetamol overdose
Seronegative hepatitis	Mitochondrial disease
Drug induced	Wilson disease
Neonatal	Autoimmune
Haemochromatosis	Hepatitis

- **Causes of chronic liver disease in older children:**

  - Post Viral hepatitis B, C
  - Autoimmune hepatitis and sclerosing cholangitis
  - Drug induced liver disease (NSAIDs)
  - Cystic fibrosis
  - Wilson disease
  - Fibropolycystic liver disease
  - Non alcoholic fatty liver disease
  - α_1 – Antitrypsin deficiency

- **Causes of prolonged (persistent) neonatal jaundice:** mcq

Prolonged Neonatal Jaundice: (>14 days age if term, >3 weeks if preterm)

Unconjugated	Conjugated (> 25 µmol/L)
Breastmilk Jaundice	Bile duct obstruction
Infection (particularly urinary tract)	Biliary atresia
Haemolytic anaemia e.g. G6PD deficiency	Choledochal cyst
Hypothyroidism	Neonatal hepatitis syndrome
High gastrointestinal obstruction	Congenital infection
Crigler – Najjar syndrome	Inborn errors of metabolism
	α_1 Antitrypsin deficiency
	Galactosaemia
	Tyrosinaemia (type 1)
	Errors of bile acid synthesis
	Progressive familial intrahepatic cholestasis
	Cystic fibrosis
	Intestinal failure associated liver disease – associated with long term parenteral nutrition

Wilson disease

(Hepato – lenticular degeneration)

- Is a genetic disease ch⁻ by defective biliary copper excretion with accumulation of toxic amounts of copper in liver – brain – kidney – cornea.
- Is AR with abnormal gene on chromosome 13
- It rare ≃ % 1: 200000

Pathogenesis:

This gene mutation leads to reduced synthesis of ceruloplasmin (the copper–binding protein) → defective copper excretion in bile → copper accumulation in liver cells → damage and release copper into circulation → accumulation in other organs (brain, kidney, and cornea).

C/P: (rarely present < 3y)

1. Hepatic (early in children) : Any form acute – chronic cirrhosis – portal HTN

2. Neurological (2nd decade) : Neuro psychiatric problems as

   - Deterioration of school performance
   - Mood and behavior changes
   - Extra pyramidal sign as tremor –incoordination – dysarthria

3. Renal

   - RTA
   - Fanconi syndrome
   - Vitamin D resistant rickets

4. Blood : Acute hemolytic anemia

5. Eyes : Kayser – Fleischer rings (not seen < 7 yr of age)



Figure 21.6 Kayser–Fleischer rings from copper in the cornea in a child with Wilson disease.

Investigation:

- Serum ceruloplasmin (low)
- Serum CU (High)
- Urinary copper (High)
- Liver biopsy (elevated hepatic copper) – confirmative *
- Genetic study : confirmative *

- Slit lamp
- LFT

Treatment:

- Penicillamine or Trientine :Both promote urinary copper excretion, reducing hepatic and central nervous system copper.
- Restrict copper containing diet
- Zinc (decrease copper absorption)
- Pyridoxine (prevent neuropathy)
- Liver transplantation : for children with acute liver failure or severe end-stage liver disease.

Congenital hepatic fibrosis:

AR disorder characterized by diffuse periportal fibrosis

C/P – Onset 1–2 year and HSM – portal HTN – Polycystic kidney (75%) – CRF

Investigation: LFT – Liver biopsy

Treatment: R_x of portal HTN & CRF

Autoimmune hepatitis and sclerosing cholangitis

- The mean age is 7 years to 10 years. It is more common in girls.
- It may present as an acute hepatitis, as fulminant hepatic failure or chronic liver disease with autoimmune features such as skin rash, arthritis, haemolytic anemia, or nephritis.
- Diagnosis is based on elevated total protein, hypergammaglobulinemia (IgG >20 g/L)/ positive autoantibodies, a low serum complement (C4)/ and typical histology. Autoimmune hepatitis may occur in isolation or in association with inflammatory bowel disease, coeliac disease, or other autoimmune diseases.
- Treatment : 90% of children with autoimmune hepatitis will respond to prednisolone and azathioprine.
- Sclerosing cholangitis is treated with ursodeoxycholic acid.

Liver tumors: Hepatoblastoma (65%) or Hepatocellular carcinoma (25%)

Hepatoblastoma:

- Unknown etiology
- Associated with FAP –BWS
- C/P : Asymptomatic abdominal mass (> Rt lobe – unifocal) ,Weight loss– Anorexia – vomiting – Abdominal pain and Jaundice are rare
- Alfa –Feto protein : High tumor marker (used for diagnosis & monitoring) mcq
- U/S Abdomen and CT or MRI.
- R_x surgery + chemotherapy

Hepatocellular carcinoma:

- Adolescents – HBV – HCV
- Prognosis in children is less satisfactory

Hepatosplenomegaly DD : mcq
1. Infections: • Viral (Hepatitis, EBV – CMV and HIV) • Bacterial (Typhoid, brucellosis and TB) • Parasitic (Malaria, bilharziasis and Toxoplasmosis) 2. Tumor: • Leukemia & lymphoma • Neuroblastoma & Wilms tumor • Hepatocellular carcinoma 3. Congenital:Cong hepatic fibrosis 4. Blood: Chronic hemolytic anemia & Osteopetrosis 5. Cardiac:Heart failure (liver) & IE (Spleen) 6. Metabolic:GSD– Galactosemia– – Gauchers – Niemann Pick & Tyrosinemia and others as wilson disease . 7. Cholestatic: Biliary atresia 8. CTD:SLE – JIA & chediak higashi disease & chronic hepatitis

Retinoblastoma mcq	Rhabdomysarcoma
Is the most common intraocular tumor in children .	Is the most common pediatric soft tissue sarcoma
Its malignant tumor of retinal cells • Sporadic – Unilateral • Hereditary – Bilateral –AD – Ch 13	It occurs more in pt with NF and Li – fraumeni syndrome
C/P: 1st 3 year • White pupillary reflex (leukocoria)** • Sequint • Hyphema 	C/P • Head and neck : 40% most common site (proptosis – nasal obst – blood stained discharge – asym mass in neck • GUS : ♀ bladder dysuria – urinary obstruction – stained vaginal discharge • Metastatic disease: 15% at diagnosis ,poor diagnosis
Investigation MRI (multi focal tumor) Fundus examination under GA.	Investigation Imaging CT – MRI bone scan Biopsy and diagnostic
Rx: Aim (Cure – preserve vision) • Enucleation is performed in advanced tumor • If Bil chemotherapy – laser therapy • Radiotherapy in advanced or recurrence • All : 1st degree relatives of children with hereditary Retinoblastoma should be screened regularly. • There increase risk of sarcoma	Rx • Preoperative chemotherapy • Resection of tumor (if possible) followed by chemo – radiotherapy • Care rate 65%

DIFFERENTIAL DIAGNOSIS OF LEUKOCORIA. mcq

1. Cataract
2. Retinoblastoma
3. Toxocariasis
4. Coat's disease
5. ROP "Retinopathy of prematurity."
6. PHPV "persistent hyperplastic primary vitreous.
7. Retinal detachment
8. Coloboma
9. Retinal dysplasia
10. Norrie's disease

Kaposi Sarcoma:

- Rare – low grade cancer
- Arises from cells of blood & lymph vessels.
- Triggered by HHV8, Common in sub Saharan Africa
- C/P : Lymphadenopathy – skin rash
- Biopsy For diagnosis.
- Chemotherapy + Antiretroviral therapy

Langerhans cell histiocytosis (LCH):

- LCH is rare disorder characterized by abnormal histiocytes (type of dendritic antigen presenting cells)
- Truly malignancy !! but sometimes aggressive behavior and response to chemotherapy

C/P

- Bone : 80% of pt (any age)
    - Single or multiple osteolytic lesion
    - Skull most common site
    - Asymptomatic / pain – local swelling
    - Diabetes insipidus may associated with skull disease , proptosis and hypothalamic infiltration
- Systemic LCH:Most aggressive
    - Skin (infancy – seborrheic rash)

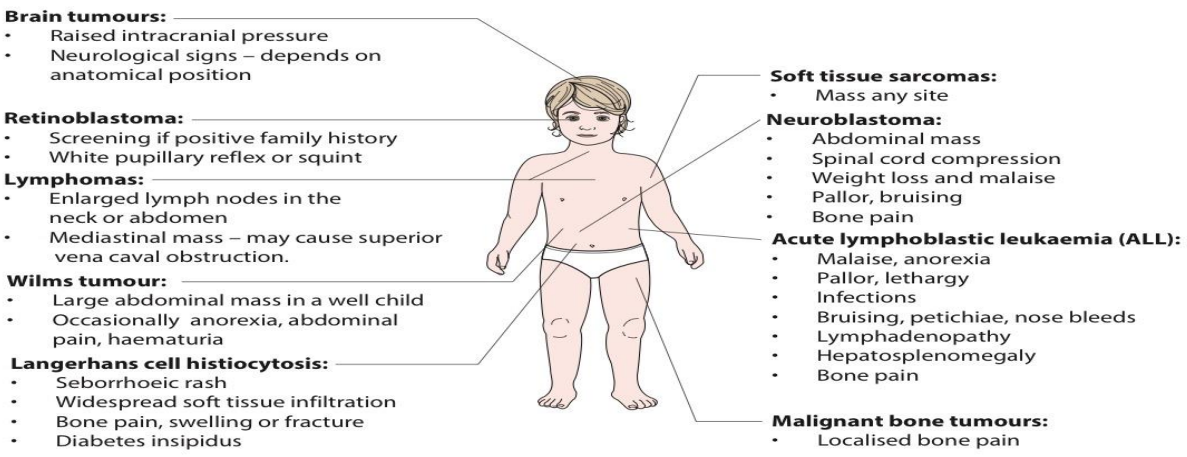
- LN – HSM – otitis media – deafness
- BM – Anemia – low plt.
- Neurological manifestation
- Usually progressive and require chemotherapy ,although spontaneous regression may occurs
- Biopsy is diagnostic (skin or bone)

Bone tumor:

1. Osteosarcoma: Is most common 1 yr malignant bone tumor in children and adolescents followed by ewing.
2. Ewing sarcoma is more common in young children < 10 years.

	Osteosarcoma mcq	Ewing sarcoma mcq
Age	2nd decade	1st decade and 2nd decade
Sex	more in ♂	more in ♂
Site	Metaphysis of long bone	Diaphysis of long bone
C/P	Local pain and swelling – H/O trauma	Local pain and swelling – fever
X Ray, CT and MRI	Sunburst pattern	Onion skinning
Mets	Lung – bone	Lung – bone
R_x	Chemo + surgery	Chemo – radio + surgery
Prognosis	without mets 70% cured With mets < 20% survival	with out mets 60% cured With mets 20–30% survival

Presentation of malignant disease in children



Pre-school (<5 years old)	School-aged	Adolescence
Acute lymphoblastic leukaemia (ALL) – peak incidence Non-Hodgkin lymphoma	Acute lymphoblastic leukaemia (ALL)	Acute lymphoblastic leukaemia (ALL) Hodgkin lymphoma
Neuroblastoma	Brain tumours	Malignant bone tumours
Wilm tumour		Soft tissue sarcomas
Retinoblastoma		

Child Abuse .mcq .case

Child abuse or maltreatment: Is physical, sexual or psychological maltreatment or neglect of a child or children especially by parents or other care giver

Abuse: refers act of commission

Neglect: to act of omission

Types:

- Neglect (34%)
- Sexual abuse (30%)
- Physical abuse (28%)
- Emotional abuse (8%)
- Fabricated/induced illness

Risk factors for child abuse :–

1. Child related:	2.Caregiver – parent related	3.Community social:
• Handicap • Prematurity • LBW • Hyperactive – different – difficult – never been well	• Alcohol – drug abuse • Mental health status • Young parental age • Step parents • Single parent • Personal H/O child abuse • Domestic violence • Unwanted pregnancy • Lack of social support • Multiple/ closely spaced births	• Poverty • Poor housing • High unemployment rate • Loss of job

Physical Abuse:

- Is any non accidental injury to child under the age of 18 year by parents or caregiver
- May include burns – human bites – beatings – shaking – strangulation – inversion in scaling water
- With resulting B viruses – burns – scars – internal injury
- Battered child syndrome
- The shaken infant

Features suggesting NAI: mcp

Injury (Exa)	History
Bruises	• Delay in seeking medical advice
• Shape :hand – object	• Explanation in compatible e⁻ injury
• Site – neck – around wrist and ankles	• Story varies
• The on the buttocks < 2yr	• Child brought by someone other than the person in presence the injury occurred
• Age cannot accurately estimated	• Unusual parent or child behavior
Burns	• Parents more interested in their own feelings and in returning home than in concern for the child
• Age not mobile child	• Abnormal parent child interaction (with drawn)
• Shape – cigarette	
• Glove or stocking burns forced immersion	
Bites	
• Shape of bite	
Features	
• Age any non mobile child	
• Site – rib – skull in young age	
• Multiple with different ages	
• Long bone in young but mobile child	

DD of NAI:

- Bruises
- Dermatological disorder as Mongolian spots can be mistaken for bruises and also bruises difficult to detect in child with dark skin (black race)
- Hematological disorders : IIP

- Burns: impetigo –SSSS.
- Fracture: ↑ osteogenesis imperfecta

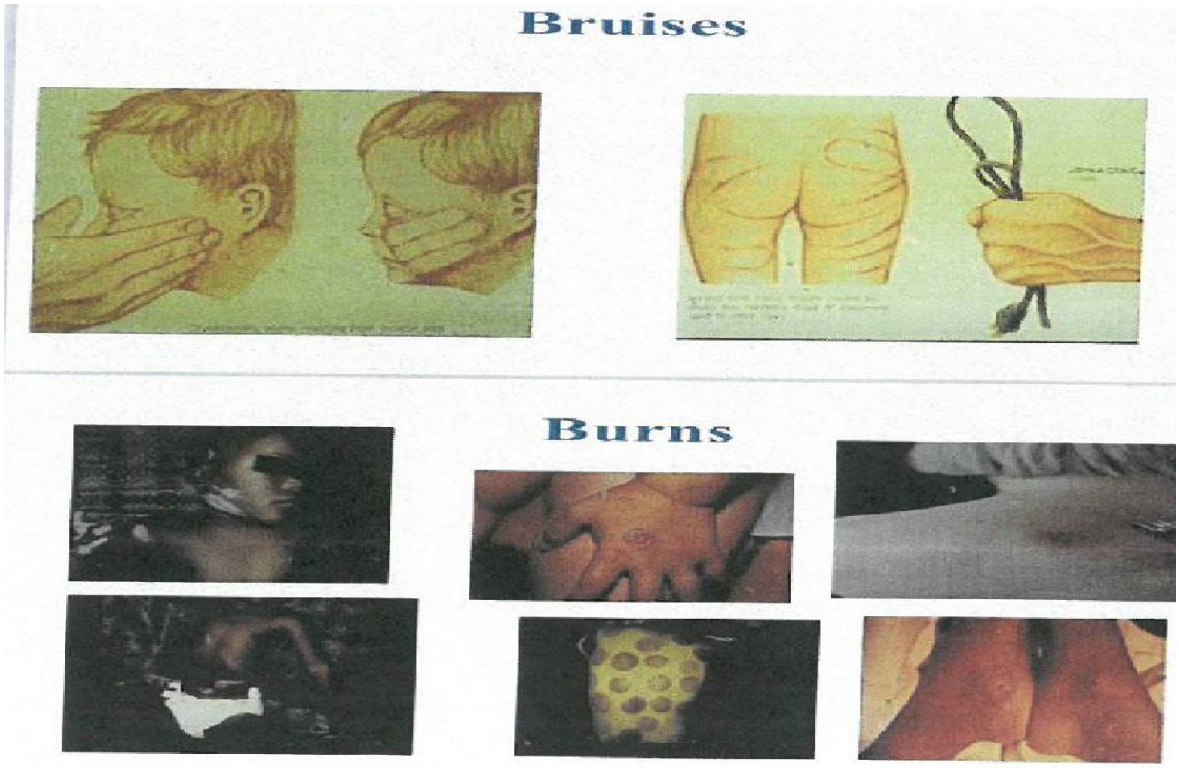
Management of NAI:

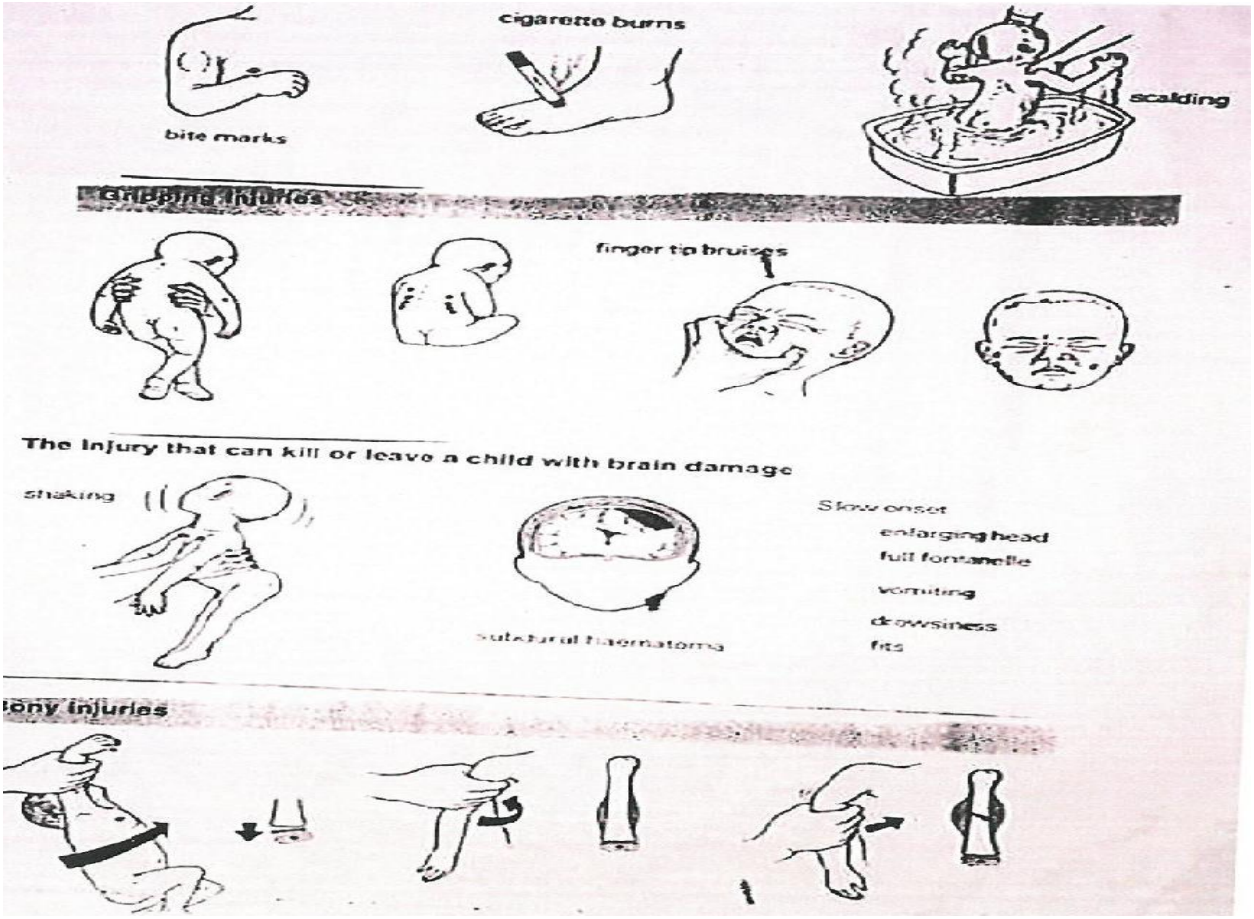
Ix

- CBC– Coagulation profile
- X ray – B/C ↑ may not detected clinically
- Full radiographic skeletal survey
- CT scan if brain injury is suspected

Rx

- Hospitalization – medical consultation
- Rx of medical propleme as ↑
- Safety of the child
- Social workers – health professional is important





Neglect:

Consider the possibility of neglect when the child

- Consistently misses imp medical appointments
- Is dirty
- Lack of needed medical dental care or immunization
- Is wearing in adequate clotting in cold weather
- Seems ravenously hungry
- Is abusing alcohol or other drugs
- Says there is no one at home to provide care

Consider the possibility of neglect when the parents or caregiver

- Seems depressed
- Appears to be indifferent to the child
- Is abusing alcohol or other drugs

Sexual Abuse:

- The child or young person may tell someone about abuse and may be pregnant and sexually transmitted disease
- Physical examine – vaginal bleeding – itching – discharge – rectal bleeding
- Behavior symptoms – Soiling – 2 year self harm – aggressive and sexual behaviors – poor school performance.

Fabricated or induced illness:

Is a group of behaviors by parents – but usually mother (>80%) which cause harm to children

1. Verbal Fabrication:

Parents fabricate symptoms and signs in the child – telling false story to doctor – leading them to believe the child is ill and require Ix and Rx

2. Induction of illness:

Suffocation of the child – poisoning – excessive intake of unnecessary substance

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References

1– Nelson textbook of pediatrics 22nd edition.

2– Illustrated textbook of paediatrics 6th edition

3– Nelson essentials of pediatrics 9th edition.

4– www.uptodate.com

5_ www.medscape.com

تنبيه:

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال ، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة

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